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Class 12
Pakistan Studies

Chapter 3
Solved Exercise

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Section 2 Geography of Pakistan

Unit 03: Land of Pakistan and Environmental Hazards

SHORT QUESTIONS

Q.1 Assess the geopolitical importance of Pakistan?

1. Pakistan lies between South Asia, Central Asia, and the Middle East, making it a key route for trade and transport.
2. It borders China, India, Iran, and Afghanistan, which are important in world politics.
3. Pakistan has a coastline on the Arabian Sea, helpful for global trade through the Gwadar Port.
4. Pakistan plays a major role in regional peace, especially in the fight against terrorism and in Afghan peace efforts.

Latitude: 24.5 to 36.5 (north latitude)
Longitude: 61-78.5 [East longitude]
18-18% area covered by Pakistan of south Asia.

Q.2 Describe the importance of Gilgit-Baltistan for tourism.

1. It has stunning mountains, rivers, and valleys like Hunza, Skardu, and Fairy Meadows.
2. Home to famous mountains like K2, the second-highest in the world, attracting climbers.
3. Rich in history, local traditions, and unique festivals that attract visitors.
4. Offers hiking, trekking, camping, and jeep safaris, which bring tourists from all over the world.

neighbouring direction;

Q.3 Why is CPEC an important project for Pakistan?

1. It will bring investment, creates jobs, and help to build industries.
2. It will help to develop roads, railways, and energy projects across the country.
3. It will connect Pakistan to all countries from East to West and North to South for trade etc.
4. It will help to reduce electricity shortage by building power plants.

Q.4 Identify the causes of earthquakes in Pakistan.

1. Pakistan lies where the Indian and Eurasian plates meet, causing pressure and quakes.
2. Many fault lines, like the Chaman Fault, run through the country and often shift.
3. The rising of the Himalayan Mountains creates stress underground, leading to earthquakes.
4. Sudden release of pressure built up in rocks causes shaking of the earth.

Q.5 With the help of a labelled diagram, explain what is Greenhouse Effect.

1. Sunlight enters Earth's atmosphere and warms the Earth's surface.
2. Earth reflects some heat back toward space as infrared radiation.
3. Greenhouse gases (like CO₂, methane, and water vapor) trap this heat.
4. This trapped heat keeps the Earth warm, but too much of it causes global warming.

Q.6 State the main causes of floods in Pakistan.

1. Strong and continuous rains overflow rivers and canals.
2. In summer, melting glaciers increase river water levels.
3. Blocked or weak drainage causes water to gather in cities and villages.
4. Cutting trees reduces the land's ability to absorb rainwater, causing floods.

Bangladesh is the only muslim country which has no border with other muslim country.

Mountains - Lesser Himalayas, Shiwalik mountains
(6000m & above) (4000 to 6000m) (below 4000m)

Additional Short Questions

Q1. Write about the total area & population of Pakistan?

Pakistan is one of the large countries of South Asia region. It has an area of 796096 square km and a population of 207,774,520 according to 2017 census.

Q2. Identify the Pakistan location of Pakistan with reference to latitude & longitude?

∴ minimum
height of
mountain & max
height of hill
is 900 metres

Pakistan lies between the latitudes of 23.45 degree north to 37.05 degree north and between the longitudes of 60.50 degree to 77.50 east. The land of Pakistan stretches over 1600km north to south and it is about 850km wide from east to west.

Q3. Identify location of Pakistan with reference to neighboring state and what area separates Pakistan from Central Asia?

Only Pakistan is in a position to provide transit trade to Afghanistan and the Central Asia States, as these countries are either landlocked or have no warm water seaports. Pakistan commands the sea lanes from oil rich Gulf States including Saudi Arabia to the Arabian Sea and most of the air traffic between east and west.

Q4. What is the difference between weather & climate?

Weather: Weather is the daily state of atmosphere in a particular area with regard to temperature, cloudiness, rainfall, wind and other meteorological conditions.

Climate: Climate is the average weather or the regular variation in weather in particular area (normally determined by the meteorologist after a study extended over 30 years).

Q5. Which factors are responsible for climate changes?

There are many factors responsible for determining the climate of an area and bringing about climatic changes i.e. location, Altitude, Air, density of the population, Industrial establishment, town planning and energy consumption behaviour of the people living in the area.

Q6. In how many zones Pakistan has been divided on the basis of temperature?

Pakistan is divided into four major zones on the basis of temperature variation:

- Highlands (Northern and North-Western Mountains Region)
- Plains (Upper and lower Indus Plains) Baluchistan Plateaus and Thar Desert:
- Coastal Areas

Q7. In how many regions of Pakistan have been divided on the basis of climate?

On the basis of climatic conditions, the land of Pakistan is divided into four major regions:

1. Sub-tropical continental highland type
2. Sub-tropical continental plateau type
3. Sub-tropical continental plain/low-land type
4. Sub-tropical coastal type *snow melts & goes to sea*

Q8. How many types of drainage system exists in Pakistan?

Pakistan's drainage system is divided into three parts:

- i. River Indus and its tributaries.
- ii Internal irrigation system
- iii. Irrigation system of the Southern Baluchistan

kareez:- underground water channels

*Latitude 23.45 - 37.5
Longitude 60.50 - 77.50*

• K-2 mountain also called mantgodwin Austin
: 6600km long Himalayas range

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Q9. What is inland Drainage System?

The drainage system covering the northern parts of the south-western Baluchistan, including Chaghi and Koh Ras mountains in the north and Siahan Mountain in the south, is called the inland drainage system.

Q10. Write name of major natural regions of Pakistan?

The land of Pakistan is divided into five main physical regions, these are: i. Plains ii. Deserts iii. Coastal Regions iv. Humid and semi-humid mountain regions v. Arid and semi- Arid Mountain Regions

• Highest peak of Himalayas range in

Q11. Write names of desert in Pakistan? i. Thal Desert (Punjab) ii. Cholistan

Desert (Punjab) iii. Thar (Nara and Thar parker) Desert (Sindh) iv. Kharan Desert (Baluchistan)

Pakistan is Nanga parbat

Q12. Into how many categories the northern mountains are divided into different categories? i. The Northern Mountains ii. Baluchistan Mountain Ranges

iii. The potwar Plateau iv. The Salt Range v. Arid and semi-Arid Mountainous regions

Q13. In how many ranges the northern mountain ranges are divided?

Ans: The ranges are further divided into four sub-ranges

- i. The Shiwalik range
- ii. The Pir Panjal Range
- iii. The Central Himalayas
- iv. The Karakoram Range

• Highest peak of Hindukush :- Tolichmir.

Q14. Which is the famous peak in the Greater Himalayan range?

Ans: Nanga Parbat is the highest and most famous peak of the central Himalayan range. The beautiful Kashmir valley is located between Pir Panjal range and Greater Himalaya range.

Q15. Define Plateau. : Plateau is the large stretch of relatively level land that is higher than the

land around it, having at least one side a steep slope falling abruptly to the lower land. A vast area of Pakistan is covered with plateaus i.e. The Potwar, Baluchistan and Salt ranges

Q16. What is water logging?

Ans: When the sub-soil water table rises so high that it comes very close to the surface level, the condition is called water-logging. This is dangerous disease of the soil as it makes agriculture almost impossible.

Q17. Define salinity and its effects?

Ans: A disease directly related to water-logging is salinity; salts present in the lower layers of the earth get dissolved in the water and water brings these salts to the surface of the earth, the water evaporates due to sun, leaving behind the salt deposits on the surface of the soil. The layers keep thickening and takes away all the fertility of the land.

Q18. Write any five causes of Deforestation.

Ans: There are several causes of deforestation. Some important reasons are given below:

1. Excessive cutting of trees.
2. Increase in salinity and water logging
3. Breaking out of fires in forests
4. Environmental pollution
5. Increase in soil erosion

→ North west mountain:-

1) Koh sulaiman 2) Koh safed hills
Swat & chitral, waziristan hills
and Kirthar mountain

Plateau: Elevation of land above sea level.

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1) doab b/w ravi & bias is called baridob.

2) doab b/w ravi & ~~chenab~~ is called sachna doab.

Q19. What is the significance of forests?

Ans: Forests play a vital role in the economy of a country; they contribute in a variety of ways. It prevents land erosion, produce timber and firewood, wood is used in the manufacturing of furniture. Wood pulp is the basic raw material for all paper industry.

Apart from it, forests are the single largest source of Oxygen production.

Q20. What are the major reasons for the scarcity of forests in our country?

Ans: Following are the major reasons that count for the scarcity of forests in our country:

1. A large part of our landform consists of dry, hilly areas; these conditions make forestations impossible or extremely difficult.
2. Water supply is insufficient
3. Majority of the population is illiterate and do not understand the importance of forests.
4. Due to fast increasing population more, land is required; forests are being cut for reclaiming more land for human settlement and more agriculture.
5. Trees are cut to meet wood and timber demand.

Q21. How to prevent deforestation?

Ans: Tree plantation is the best way to prevent deforestation, prevent land erosion, improve atmosphere and increase rainfall. Tube-wells gradually bring the underground salts to the surface of the soil, these salts prevent plant growth.

Q23. Name some major air pollutants.

Ans: some major air pollutants are:

1. Oil and fuel
2. Coal
3. Atomic devices
4. Dangerous chemicals.
5. Radiations emitted from high-tension electric wires
6. Pollen grains.

Q24. Define Water pollutant.

Ans: Water is the second largest source of pollution. Water pollutes due to micro-organisms for example the germs which cause diseases like Hepatitis etc. organic pollution for example sewerage in the urban localities, pesticides, fertilizers and animals refuse. Similarly, sedimented pollution provides favorable environment of sub-water plants which is difficult to filter and damages the kidneys.

Q25. What is Land/Soil pollution?

Ans: Land or soil is polluted in a number of ways, e.g. indecent town planning, Cutting of forests, Absorption of poisonous chemicals, herbicides, pesticides, and insecticides into the soil through water. The Overgrazing of pastures and unnecessary burning and cutting of plants causes Land or Soil pollution.

Q26. What are the measures to control environmental pollution?

Ans: We should adopt the following measures to prevent environmental pollutions. Following are some of the measure

1. Creation of general awareness

2. Population planning
3. Prevention of Urbanization
4. Plantation
5. Conservation of Resources.

Q27. Point out the environmental hazards for Human life.

Ans: Following are some of the hazards caused for human life due to environmental pollution:

1. Deformities of the newborn children
2. Kidney disorders.
3. Disorder of digestive and nervous system
4. Deforestation and desertification
5. Loss of marine life
6. Reductions of the agriculture produce.

LONG QUESTIONS

Q.1 Analyse the strategic importance of Pakistan for its neighbouring countries.

Strategic Importance of Pakistan for Its Neighbouring Countries

1. Geographical Location

Pakistan is located at the crossroads of South Asia, Central Asia, and the Middle East. This location makes it a bridge between energy-rich countries like Iran and Central Asia, and energy-deficient countries like India and China. Its position offers access to the Arabian Sea, making it a key transit point for global trade routes.

2. China-Pakistan Economic Corridor (CPEC)

China values Pakistan greatly due to the CPEC, which connects China's western region to the Gwadar Port in Pakistan. This reduces China's trade distance and offers a secure trade route that bypasses the South China Sea. It also helps China expand its Belt and Road Initiative (BRI).

3. Access to Warm Waters

Landlocked Central Asian states, such as Uzbekistan and Turkmenistan, depend on Pakistan for potential access to warm waters via its ports, especially Gwadar. This access could boost their trade and energy exports globally.

4. Defence and Military Cooperation

Pakistan has strong defence ties with China, and military cooperation with Iran and Afghanistan. Its large and capable military, nuclear power status, and regional influence make it a strategic partner in regional security matters.

5. Counter-Terrorism Role

Pakistan plays an important role in the fight against terrorism. It shares borders with Afghanistan, where many terrorist groups are active. Stability in Pakistan helps improve security for all neighbouring countries.

6. Gateway to Central Asia

Pakistan serves as a gateway for trade and energy projects from Central Asia to South Asia and beyond. This makes it valuable for regional integration and economic cooperation.

7. Cultural and Religious Links

Pakistan shares deep cultural, religious, and linguistic ties with Iran, Afghanistan, and India. These commonalities strengthen people-to-people connections and diplomatic dialogue in the region.

8. Economic Trade Routes

Through road and rail networks, Pakistan offers vital trade routes to India (if political ties improve), China (via CPEC), and Afghanistan. Its location supports regional trade and connectivity.

9. Energy Transit Country

Pakistan can act as an energy corridor, transporting gas and oil from Iran and Central Asia to other regions. Projects like the Iran-Pakistan gas pipeline show its importance in energy supply chains.

10. Influence in Muslim World

As a major Muslim country with a powerful military, Pakistan holds influence in the Muslim world. Countries like Iran and Afghanistan consider Pakistan a significant voice in regional Islamic affairs.

11. Strategic Buffer Zone

For China and Iran, Pakistan acts as a buffer state against instability in Afghanistan and tensions with India. It helps prevent conflicts from spreading and provides a safer zone near their borders.

12. Humanitarian and Refugee Role

Pakistan has hosted millions of Afghan refugees over the decades, showing its humanitarian importance. Its cooperation in dealing with migration and displaced populations is vital for peace and development in the region.

Conclusion

Pakistan holds strategic importance for its neighbours due to its **unique geographical location, strong defence, economic corridors, and deep cultural ties**. It acts as a bridge between regions, a trade and energy hub, a security partner, and a cultural connector. The country's policies and stability directly affect the peace, economy, and development of the entire region. Therefore, for countries like China, Afghanistan, Iran, India, and Central Asian states, Pakistan remains a **key regional player with immense strategic value**.

Q.2 Evaluate the importance of Gwadar port with reference to the economic progress of Baluchistan.

Importance of Gwadar Port for the Economic Progress of Baluchistan

1. Gateway to Global Trade

Gwadar Port is located on the south-western coast of Baluchistan, near the Persian Gulf. Its deep-sea structure and ideal location make it a major gateway for international trade, especially for countries like China, Afghanistan, and Central Asian states.

2. CPEC and Regional Connectivity

Gwadar Port is the backbone of the **China-Pakistan Economic Corridor (CPEC)**. This mega project connects western China to Gwadar through road and rail networks, boosting regional connectivity and putting Baluchistan on the global trade map.

3. Job Creation for Local People

The development of the port and surrounding infrastructure creates thousands of direct and indirect jobs. These include positions in construction, transportation, security, trade, shipping, and services, benefiting the people of Baluchistan.

4. Growth of Local Businesses

As trade through Gwadar increases, local businesses like hotels, shops, logistics, and services are expanding. This brings economic activity and helps reduce poverty in surrounding areas.

5. Improvement in Infrastructure

To support Gwadar Port, new highways, roads, airports, and railway lines are being built. These infrastructure projects improve transportation, connect remote areas, and increase investment opportunities in Baluchistan.

6. Development of Special Economic Zones (SEZs)

Several **Special Economic Zones (SEZs)** are being planned near Gwadar. These zones offer tax breaks and facilities to local and foreign investors, attracting industries and encouraging economic development in the province.

7. Foreign Investment Opportunities

Gwadar is drawing international investors, especially from China and Gulf countries. Investments in shipping, energy, trade, and tourism bring capital into Baluchistan and improve its financial condition.

8. Boost to Tourism and Real Estate

Gwadar's scenic beauty and peaceful coastline make it a potential tourist destination. The port's development is also raising the value of real estate, opening doors for hotels, resorts, and housing projects.

9. Fisheries and Marine Industry Growth

The port also benefits the local fishing industry, which is a major livelihood in Baluchistan. Better storage, transport, and export facilities help fishermen earn more and expand their businesses.

10. Education and Skill Development

To meet the demands of the growing port city, technical institutes and training centers are being set up. These facilities aim to equip local youth with modern skills for jobs in trade, transport, engineering, and management.

11. Reduction of Regional Disparities

Baluchistan has long been the least developed province of Pakistan. Gwadar's development promises to reduce the economic gap between Baluchistan and other provinces by bringing in resources, development, and government attention.

12. Strategic Economic Position

Gwadar Port not only strengthens Baluchistan but also enhances Pakistan's position in global maritime trade. Baluchistan can become a hub of economic activity if the port is managed well with local participation.

Conclusion

Gwadar Port is a game-changer for Baluchistan's economy. It has the potential to turn the province into a center of **trade, employment, tourism, and industry**. With better policies, security, and local involvement, Gwadar can bring long-term benefits not just for Baluchistan but for the entire country. The port is a **golden opportunity** to uplift the lives of the people of Baluchistan and bring lasting peace and prosperity to the region.

Q.3 The importance of tourist potential of Pakistan to its unique geographical location.

1. Introduction

Pakistan is a country blessed with a rich blend of landscapes, cultures, and histories. Its unique geographical location makes it a land of enormous tourist potential. From towering mountains in the north to golden deserts in the south, Pakistan's natural and historical wonders attract tourists worldwide.

2. Strategic Location at the Crossroads

Pakistan lies at the crossroads of South Asia, Central Asia, and the Middle East. This location makes it a connecting route for travellers and traders for centuries. It gives Pakistan a unique position to attract tourists from different regions easily.

3. The Mighty Northern Mountains

The northern areas of Pakistan boast some of the highest peaks in the world, including K2, the second tallest mountain. These ranges offer thrilling experiences such as hiking, trekking, mountaineering, and skiing, drawing adventure lovers from across the globe.

4. Cultural Diversity

Pakistan's diverse ethnic groups, languages, and customs offer a cultural richness that few countries can match. From Pashtun hospitality in Khyber Pakhtunkhwa to the colorful traditions of Punjab and Sindh, tourists experience a wide variety of cultures in one trip.

5. Rich Historical Heritage

Pakistan is home to ancient civilizations such as the Indus Valley Civilization, Mohenjo-Daro, and Harappa. Historical cities like Taxila, Lahore, and Multan have magnificent architecture, forts, shrines, and museums that reflect centuries of history.

6. Religious Tourism

Pakistan hosts several religious sites important to Muslims, Hindus, Sikhs, and Buddhists. The Kartarpur Corridor has opened doors for Sikh pilgrims worldwide, while Buddhist ruins in Taxila attract those following the Buddha's path.

7. Breathtaking Natural Beauty

From the serene lakes of Skardu to the green valleys of Swat and Hunza, Pakistan offers jaw-dropping natural beauty. Tourists find peace and refreshment in these naturally gifted locations, perfect for eco-tourism.

8. Variety of Climates

Due to its geography, Pakistan experiences all four seasons. Tourists can enjoy snow-covered peaks in winter, vibrant spring flowers, summer adventures in the mountains, and golden autumns—all within the same country.

9. Unique Wildlife and National Parks

Pakistan's diverse habitats are home to rare species such as the snow leopard, markhor, and Himalayan brown bear. National parks like Khunjerab and Hingol attract wildlife enthusiasts and photographers.

10. Coastal and Desert Attractions

The coastline of Baluchistan offers hidden beaches like Kund Malir and Astola Island. The Thar Desert in Sindh offers desert safaris, cultural festivals, and a unique desert lifestyle experience.

11. Accessibility to Neighboring Regions

Pakistan shares borders with China, India, Iran, and Afghanistan, which makes it a gateway for regional tourism. The CPEC route and new border infrastructure improve accessibility for tourists from nearby countries.

12. Economic and Image Boost: Tourism can play a huge role in improving Pakistan's economy and international image. As more tourists visit, they help create jobs, promote local crafts, and spread a positive image of the country abroad.

Conclusion

Pakistan's unique geographical location gives it a rare advantage in the global tourism industry. With its blend of natural wonders, rich history, diverse cultures, and strategic position, Pakistan holds great potential as a top tourist destination. With better infrastructure, safety, and promotion, it can become a must-visit country for travelers worldwide.

Q.4 Examine the importance of the National Disaster Management Authority (NDMA) in managing the natural disaster in Pakistan.

1. Introduction

Pakistan faces multiple natural and man-made disasters, such as floods, earthquakes, droughts, and more. To manage these challenges, the **National Disaster Management Authority (NDMA)** plays a central role in disaster planning, response, and recovery across the country.

2. Foundation of NDMA

NDMA was established in 2005 after the devastating earthquake that exposed serious gaps in disaster preparedness. It operates under the **National Disaster Management Act, 2010** and serves as the main federal authority to manage disaster risk reduction and emergency response.

3. Central Coordinating Body

NDMA coordinates between federal, provincial, and local agencies. It ensures that all parts of the government and relevant organizations work together before, during, and after a disaster to save lives and reduce losses.

4. Early Warning Systems

NDMA plays a vital role in developing and sharing **early warning systems** for potential hazards such as floods and cyclones. These warnings help communities prepare in advance and avoid heavy damage.

5. Disaster Preparedness Plans

One of NDMA's key tasks is preparing **disaster management plans**. These include national, provincial, and district-level plans that focus on risk assessment, evacuation strategies, and resource allocation.

6. Emergency Response Operations

When a disaster strikes, NDMA takes charge of the **emergency response**. It deploys rescue teams, coordinates with the army, arranges medical aid, and ensures timely delivery of relief supplies such as food, tents, and medicines.

7. Training and Capacity Building

NDMA conducts training programs for government officials, rescue workers, and volunteers. This builds local capacity to respond effectively and strengthens community resilience.

8. Role in Natural Disasters: NDMA has played a significant role in managing major disasters like the 2005 earthquake, 2010 floods, and more recently, glacial lake outburst floods and forest fires. Their coordinated efforts reduce human and material losses.

9. Public Awareness Campaigns: NDMA runs awareness campaigns to educate the public about disaster risks and safety measures. School programs, media outreach, and public drills help communities understand how to act during emergencies.

10. Collaboration with International Agencies

NDMA works with global partners such as the United Nations, Red Crescent, World Bank, and other donor agencies. This cooperation brings technical expertise, funding, and modern equipment to Pakistan's disaster management system.

11. Role in Climate Change Preparedness

As climate change increases the frequency of natural disasters, NDMA plays an important part in developing **climate resilience strategies**, including water management, forest protection, and urban planning.

12. Data Collection and Research

NDMA gathers and analyzes **disaster data** to support better planning and decision-making. This includes hazard maps, risk profiles, and historical trends, which help reduce vulnerabilities in disaster-prone areas.

Conclusion

The National Disaster Management Authority (NDMA) is a cornerstone of Pakistan's disaster management system. From planning and early warning to response and recovery, NDMA's efforts protect millions of lives and help the country recover from major disasters. As risks grow due to climate change and urbanization, NDMA's role will become even more important in building a safer, more resilient Pakistan.

Q.5 Explain with reference to examples. The effects of floods on the environment and human life.

1. Introduction

Floods are among the most common and destructive natural disasters. They occur when water overflows onto normally dry land, often due to heavy rainfall, river overflow, dam failure, or glacier melting. In countries like Pakistan, floods have serious effects on both the environment and human life.

2. Damage to Homes and Infrastructure

Floods can destroy houses, roads, bridges, schools, and hospitals. For example, the **2022 floods in Pakistan** damaged over **2 million homes** and thousands of kilometers of roads, leaving millions homeless and disconnected from basic services.

3. Loss of Human Lives

Floods often result in the loss of lives due to drowning, electrocution, or building collapse. In 2010, floods in Pakistan killed around **2,000 people**, showing the deadly impact of these disasters.

4. Spread of Diseases

Standing floodwater becomes a breeding ground for mosquitoes and other insects, increasing the risk of diseases like **malaria, dengue, and cholera**. Poor sanitation and contaminated water also lead to widespread illness.

5. Destruction of Agriculture

Floods ruin crops, wash away fertile soil, and damage irrigation systems. In Pakistan, the 2022 floods destroyed over **3 million acres** of crops, causing food shortages and rising prices for basic items like wheat, rice, and vegetables.

6. Loss of Livelihood

Many people in rural areas depend on farming, fishing, or small businesses. Floods destroy their income sources. After the 2022 floods, thousands of Pakistani farmers lost their only source of earning, pushing them into poverty.

7. Environmental Pollution

Floodwater carries garbage, chemicals, sewage, and oil, polluting rivers, lakes, and groundwater. This not only harms aquatic life but also affects humans who depend on these water sources.

8. Soil Erosion and Land Degradation

Strong flood currents wash away topsoil, which is the most fertile layer. This leads to **soil erosion** and makes the land less productive for future farming, especially in flood-prone areas like Sindh and South Punjab.

9. Damage to Wildlife and Ecosystems

Floods can destroy animal habitats, drown wildlife, and disturb natural ecosystems. For example, in the **Indus Delta**, floods have affected mangrove forests and threatened fish and bird species.

10. Forced Migration and Displacement

Floods force people to leave their homes, creating **climate refugees**. In 2022, more than **8 million people** were displaced across Pakistan, many of whom had to live in temporary camps without proper food or shelter.

11. Impact on Education

Schools are often used as shelters or are damaged by floodwaters, stopping children's education. In recent floods, over **27,000 schools** were damaged or destroyed, interrupting the studies of millions of children.

12. Psychological Stress and Trauma

Losing homes, family members, or livelihoods can cause deep emotional stress, especially for children and women. Many survivors suffer from **anxiety, depression, and post-traumatic stress** after experiencing a major flood.

Conclusion

Floods have far-reaching effects on both the environment and human life. They destroy homes, farmland, ecosystems, and communities. The example of Pakistan's 2010 and 2022 floods shows how serious the consequences can be. To reduce the damage, there must be better flood control systems, disaster preparedness, and public awareness to protect both people and the planet.

Q.6 What factors affect the climate change in Pakistan Elaborate with examples.

Factors Affecting Climate Change in Pakistan

1. Deforestation

One of the major causes of climate change in Pakistan is deforestation. Forests act as the planet's lungs, absorbing carbon dioxide (CO₂) from the air. However, in Pakistan, large areas of trees are cut down for agriculture, urbanization, and wood fuel. This reduces the number of trees that can absorb CO₂, leading to increased greenhouse gases in the atmosphere. For example, the deforestation in the northern areas like Chitral and Gilgit-Baltistan has contributed to changes in local climate and increased vulnerability to floods.

2. Industrial Emissions

Factories and industries release significant amounts of greenhouse gases, especially carbon dioxide, methane, and nitrous oxide. In Pakistan, industries like cement, textile, and power generation contribute heavily to air pollution. For example, the brick kilns in urban and rural areas emit large amounts of smoke and gases, adding to global warming.

3. Use of Fossil Fuels

Pakistan heavily relies on fossil fuels such as coal, oil, and natural gas for energy. Burning these fuels releases large quantities of CO₂, a leading greenhouse gas that traps heat in the atmosphere. The energy sector is one of the largest contributors to climate change in Pakistan, and continued dependence on fossil fuels makes the problem worse.

4. Urbanization

Rapid urbanization in Pakistan has led to increased construction, transportation, and energy consumption. Cities like Karachi, Lahore, and Islamabad are expanding quickly, leading to more emissions from vehicles, construction machinery, and electricity use. For example, traffic congestion in Lahore results in high emissions of pollutants which contribute to climate change.

5. Agriculture and Livestock

Agriculture is a vital part of Pakistan's economy but also a significant source of methane and nitrous oxide emissions—powerful greenhouse gases. Over-irrigation, use of chemical fertilizers, and raising large numbers of livestock release these gases. For instance, rice paddies in Punjab produce methane, which exacerbates global warming.

6. Water Management and Glacial Melting

Pakistan's glaciers in the Himalayas and Hindu Kush are melting due to rising temperatures. This affects water availability for millions of people. Melting glaciers also contribute to rising water levels in rivers, increasing the risk of floods, which further impacts climate stability.

7. Climate-Resilient Agriculture Challenges

Changing weather patterns, unpredictable rainfall, and increased droughts make agriculture more difficult in Pakistan. These challenges are linked to climate change and threaten food security, affecting millions who depend on farming.

8. Melting of the Himalayas and Coastal Erosion

The melting of glaciers in the Himalayas, often called the "Third Pole," leads to rising sea levels and coastal erosion. Pakistani coastal areas like Karachi face increased flooding and loss of land, which impacts local communities and biodiversity.

9. Natural Disasters

Pakistan faces frequent natural disasters such as floods, droughts, and heatwaves, all of which are intensified by climate change. For example, the devastating 2010 floods affected approximately 20 million people, primarily caused by excessive rainfall linked to changing climate patterns.

10. Lack of Environmental Awareness

Many people in Pakistan are not aware of the environmental impact of their actions. This lack of awareness leads to practices like illegal logging, burning waste, and excessive water use, which worsen climate change.

11. Inadequate Policy Implementation

Though Pakistan has policies for climate change mitigation, enforcement is weak. Limited resources and political will prevent effective measures from being implemented fully, hampering efforts to control climate change.

12. Poverty and Socioeconomic Factors

Poverty restricts people's ability to adopt environmentally friendly practices. Many poor communities depend on wood fuel for cooking and heating, which increases deforestation and pollution.

Conclusion

In conclusion, climate change in Pakistan is driven by a combination of environmental, economic, and social factors. Deforestation, reliance on fossil fuels, industrial emissions, and rapid urbanization are major contributors. Melting glaciers, water management issues, and natural disasters further worsen the situation. Despite challenges, raising awareness, improving policies, and adopting cleaner technologies are essential steps to mitigate climate change. Pakistan must work collectively—government, communities, and individuals—to protect its environment for future generations and reduce the adverse effects of climate change.

Section 2 Geography of Pakistan